

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain the various engineering materials necessary for fabrication of induction motor. (CO2)
- Q.24 Differentiate between conductors, semiconductors, and insulators with the help of energy band diagrams. (CO1).
- Q.25 Discuss low resistivity conducting material with their properties and applications (CO1)

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3rd Sem / Electrical, Power Station Engg., Elect. & Eltx. Engg., Fire Tech & Safety

Subject : Electrical Engineering Materials / Electrical and Electronics Engineering Materials

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 The temperature coefficient of resistance is negative for. (CO1)
- a) Mercury b) Steel
c) Carbon d) Copper
- Q.2 The most used conducting material is _____ (CO1)
- a) Gold b) Copper
c) Silver d) Iron
- Q.3 Which of the following material is used for making the permanent magnet. (CO3)
- a) Alnico b) Mu-metal
c) Copper d) Nickel Iron

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Q.4 By laminating the stator core of 3-phase induction motor, which out of the following losses are reduced (CO2)

- a) Hysteresis losses b) Eddy current losses
- c) Frictional losses d) copper losses

Q.5 In case of Insulator, valence band and conduction band have (CO1)

- a) Less gap b) Moderate gap
- c) Large gap d) Overlapped

Q.6 Thermocouple works on the principle of _____ (CO3)

- a) Joule effect b) Pettier effect
- c) Thomson effect d) Seebach effect

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Write any two properties of teflon. (CO1)

Q.8 Air is an example of insulating material. (True/False) (CO1)

Q.9 Name any two hard magnetic materials. (CO3)

Q.10 Define Doping. (CO1)

Q.11 Brass is an alloy of _____ and _____. (CO2)

Q.12 Give the full form of ACSR. (CO1)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain the thermal classification of insulating material. (CO1)

Q.14 Write the properties of bakelite and its two uses. (CO1)

Q.15 Write properties of Mu metal (CO3)

Q.16 Write the properties and applications of constantan. (CO1)

Q.17 Write a short note on Bi-metals and its applications. (CO3)

Q.18 Give comparison between soft and hard magnetic materials. (CO3)

Q.19 Explain in brief about the thermosetting material. (CO1)

Q.20 Define super conductivity with two examples and Mention its applications. (CO1)

Q.21 Define Hysteresis losses and discuss the factors on which hysteresis loss depends. (CO2)

Q.22 Define semiconductor. Discuss its engineering applications. (CO1)

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